

POSITIVITY AND SOLVABILITY OF THE SCALAR DEFORMED HERMITIAN–YANG–MILLS EQUATION

ABSTRACT. In the positive supercritical range, smooth scalar deformed Hermitian–Yang–Mills solvability is characterized by a shifted Kähler class on surfaces, by subsolutions analytically, and by intersection inequalities along a ray on compact Kähler manifolds. For projective manifolds, only the endpoint inequalities are needed. In the stated hypercritical range, properness is another equivalent criterion. We give the distinct hypotheses for three scalar flows, separate special Lagrangian existence from smooth-flow convergence in the Thomas–Yau problem, and state a toric large-volume correspondence. On surfaces, stability under every integral scaling allows a nef boundary class and therefore does not imply smooth dHYM solvability.

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1. INTRODUCTION

1.1. The equation and its branch. Let X be a compact Kähler manifold of dimension $n \geq 2$. Fix a real class $A \in H^{1,1}(X, \mathbb{R})$, a Kähler class $\Omega \in \mathcal{K}_X$, and representatives $\alpha \in A$ and $\omega \in \Omega$, with α smooth and closed and $\omega > 0$. For real smooth potentials, put

$$\alpha_\varphi = \alpha + \sqrt{-1}\partial\bar{\partial}\varphi, \quad \omega_\psi = \omega + \sqrt{-1}\partial\bar{\partial}\psi > 0. \quad (1.1)$$

For a Kähler form χ and a real $(1, 1)$ -form η , let $\lambda_j(\eta; \chi)$ be the eigenvalues of η relative to χ . The phase and its determinant identity are

$$\vartheta_\chi(\eta) = \sum_{j=1}^n \arctan \lambda_j(\eta; \chi), \quad \arctan \lambda_j \in (-\pi/2, \pi/2), \quad (1.2)$$

$$(\chi + \sqrt{-1}\eta)^n = e^{\sqrt{-1}\vartheta_\chi(\eta)} \prod_{j=1}^n \sqrt{1 + \lambda_j(\eta; \chi)^2} \chi^n. \quad (1.3)$$

Fix a *supercritical* target $\hat{\theta} \in ((n-2)\pi/2, n\pi/2)$. The scalar deformed Hermitian–Yang–Mills equation asks that this real phase be constant:

- For fixed ω , the unknown is $\varphi \in C^\infty(X, \mathbb{R})$, and

$$\vartheta_\omega(\alpha_\varphi) = \hat{\theta}. \quad (1.4)$$

- With only (A, Ω) fixed, the unknown is $(\varphi, \psi) \in C^\infty(X, \mathbb{R})^2$, and

$$\vartheta_{\omega_\psi}(\alpha_\varphi) = \hat{\theta}, \quad \omega_\psi > 0. \quad (1.5)$$

Set

$$Z_X(A, \Omega) = \langle (\Omega + \sqrt{-1}A)^n, [X] \rangle. \quad (1.6)$$

A solution satisfies

$$e^{-\sqrt{-1}\hat{\theta}} Z_X(A, \Omega) = \int_X \prod_{j=1}^n \sqrt{1 + \lambda_j(\alpha_\varphi; \omega_\psi)^2} \omega_\psi^n > 0, \quad (1.7)$$

with $\psi = 0$ for a fixed background. We call this the compatibility condition. It is imposed wherever a compatible target is specified.

1.2. Existence: surfaces, subsolutions and numerical tests.

Theorem 1.1 (Jacob–Yau [JY17, Theorem 1.2]). *Let X be a compact Kähler surface, $\Omega \in \mathcal{K}_X$ and $A \in H^{1,1}(X, \mathbb{R})$ with $\langle A \smile \Omega, [X] \rangle > 0$. Set*

$$\hat{\theta} = \text{Arg}(\langle (\Omega + \sqrt{-1}A)^2, [X] \rangle) \in (0, \pi), \quad c = \cot \hat{\theta}.$$

The following are equivalent:

- (i) *For some, equivalently every, Kähler form $\omega \in \Omega$, there is a smooth closed real $(1, 1)$ -form $\eta \in A$ with $\vartheta_\omega(\eta) = \hat{\theta}$.*
- (ii) *$A + c\Omega \in \mathcal{K}_X$.*
- (iii) *$\langle A + c\Omega, [C] \rangle > 0$ for every irreducible curve $C \subset X$.*

The proof in Section 3 reduces the equation to

$$\beta_\varphi = \alpha_\varphi + c\omega > 0, \quad \beta_\varphi^2 = (1 + c^2)\omega^2.$$

For a fixed ω and target $\hat{\theta}$, a smooth potential φ is a *C-subsolution* if its eigenvalues $\mu_j = \lambda_j(\alpha + \sqrt{-1}\partial\bar{\partial}\varphi; \omega)$ satisfy

$$\sum_{j \neq k} \arctan \mu_j > \hat{\theta} - \frac{\pi}{2}, \quad k = 1, \dots, n, \quad x \in X. \quad (1.8)$$

Theorem 1.2 (Lin, preprint [Lin23, Theorem 1.5]). *Let (X, ω) be a compact Kähler n -fold with $n \geq 2$, and let α be a smooth closed real $(1, 1)$ -form. Fix $\hat{\theta} \in ((n-2)\pi/2, n\pi/2)$. Assume both of the following:*

- $e^{-\sqrt{-1}\hat{\theta}} \int_X (\omega + \sqrt{-1}\alpha)^n \in \mathbb{R}_{>0}$.
- *There is a smooth C-subsolution $\varphi \in C^\infty(X, \mathbb{R})$ for the target $\hat{\theta}$.*

Then $\vartheta_\omega(\alpha + \sqrt{-1}\partial\bar{\partial}\varphi) = \hat{\theta}$ has a smooth solution, unique modulo constants.

This is the formulation in the cited preprint. The published Collins–Jacob–Yau theorem also assumes $\vartheta_\omega(\alpha_\varphi) > (n-2)\pi/2$; Section 4 distinguishes these hypotheses and gives the estimates [CJY20, Theorem 1.2].

For the numerical tests, use

$$\theta = \frac{n\pi}{2} - \hat{\theta} \in (0, \pi), \quad Q_\omega(\eta) = \frac{n\pi}{2} - \vartheta_\omega(\eta), \quad (1.9)$$

$$\mathcal{F}_p(A, \Omega; \theta) = \operatorname{Re}(A + \sqrt{-1}\Omega)^p - \cot \theta \operatorname{Im}(A + \sqrt{-1}\Omega)^p. \quad (1.10)$$

Thus $Q_\omega(\eta) = \theta$ is the same equation with the complementary phase.

Theorem 1.3 (Chu–Lee–Takahashi [CLT24, Theorem 1.3]). *Let X be a compact Kähler n -fold with $n \geq 2$, let $A \in H^{1,1}(X, \mathbb{R})$ and let $\omega \in \Omega$ be a Kähler form. Fix $\theta \in (0, \pi)$ and assume $\langle \mathcal{F}_n(A, \Omega; \theta), [X] \rangle = 0$. Then there is a smooth closed real $(1, 1)$ -form $\eta \in A$ satisfying $Q_\omega(\eta) = \theta$ if and only if, for every $t \geq 0$ and every irreducible analytic subvariety $V \subseteq X$ of dimension $1 \leq p \leq n$,*

$$\langle \mathcal{F}_p(A + t\Omega, \Omega; \theta), [V] \rangle \begin{cases} > 0, & p < n, \\ \geq 0, & p = n. \end{cases}$$

Theorem 1.4 (Chu–Lee–Takahashi [CLT24, Corollary 1.5]). *Let X be smooth projective of dimension $n \geq 2$, let $A \in H^{1,1}(X, \mathbb{R})$ and let $\omega \in \Omega$ be a Kähler form. Fix $\theta \in (0, \pi)$. Then there is a smooth closed real $(1, 1)$ -form $\eta \in A$ satisfying $Q_\omega(\eta) = \theta$ if and only if both of the following hold:*

- $\langle \mathcal{F}_n(A, \Omega; \theta), [X] \rangle = 0$.
- $\langle \mathcal{F}_p(A, \Omega; \theta), [V] \rangle > 0$ for every irreducible proper analytic subvariety $V \subset X$ with $1 \leq p = \dim V < n$.

The first criterion is cohomological, so the fixed- and variable-background existence problems are equivalent for a prescribed pair (A, Ω) . Projectivity removes the tests with $t > 0$; on a general Kähler manifold, endpoint positivity alone can miss the real phase branch (Example 5.1).

1.3. Properness and the choice of scalar flow. Fix ω, α and a compatible target $\hat{\theta}$. Define the almost-calibrated potentials and the phase densities by

$$\begin{aligned} \mathcal{H} &= \{\varphi \in C^\infty(X, \mathbb{R}) : |\vartheta_\omega(\alpha_\varphi) - \hat{\theta}| < \pi/2\}, \\ \rho_\varphi &= \operatorname{Re}(e^{-\sqrt{-1}\hat{\theta}}(\omega + \sqrt{-1}\alpha_\varphi)^n) > 0, \\ \sigma_\varphi &= \operatorname{Im}(e^{-\sqrt{-1}\hat{\theta}}(\omega + \sqrt{-1}\alpha_\varphi)^n). \end{aligned}$$

When $0 \in \mathcal{H}$, the functional $\mathcal{J}$ and the path-length metric d_1 are specified by

$$d\mathcal{J}_\varphi(v) = - \int_X v \sigma_\varphi, \quad \mathcal{J}(0) = 0, \quad \|v\|_{1,\varphi} = \int_X |v| \rho_\varphi.$$

Here d_1 is the infimum of $\int_0^1 \|\dot{\varphi}_t\|_{1,\varphi_t} dt$ over smooth paths in $\mathcal{H}$ joining the two endpoints. For $0 < \theta < \pi/2$, put

$$\mathcal{H}_\varepsilon = \{\varphi \in \mathcal{H} : Q_\omega(\alpha_\varphi) > \varepsilon\}, \quad 0 < \varepsilon < \theta.$$

Theorem 1.5 (Chu–Lee [CL26, Theorem 1.4]). *Let (X, ω) be a compact Kähler n -fold with $n \geq 4$, and let α be a smooth closed real $(1, 1)$ -form. Fix $0 < \theta < \pi/2$ and $\widehat{\theta} = n\pi/2 - \theta$. Assume*

- $e^{-\sqrt{-1}\widehat{\theta}} \int_X (\omega + \sqrt{-1}\alpha)^n \in \mathbb{R}_{>0}$;
- $0 \in \mathcal{H}$.

Then $Q_\omega(\alpha_\varphi) = \theta$ is smoothly solvable if and only if, for every $\varepsilon \in (0, \theta)$, there exist $\delta_\varepsilon, C_\varepsilon > 0$ such that

$$\mathcal{J}(\varphi) \geq \delta_\varepsilon d_1(\varphi, 0) - C_\varepsilon \quad \text{for all } \varphi \in \mathcal{H}_\varepsilon \text{ with } \sup_X \varphi = 0.$$

The constants may depend on ε ; the statement does not impose one coercivity constant on all of $\mathcal{H}$.

A target is *hypercritical* when $\widehat{\theta} > (n-1)\pi/2$, equivalently $0 < \theta < \pi/2$; an initial potential is hypercritical when its pointwise phase exceeds $(n-1)\pi/2$.

For $n \geq 2$, with the same fixed data and compatible target, the three scalar equations have different initial-data conclusions:

- (i) If $0 < \theta < \pi/2$ and $\varphi_0 \in \mathcal{H}$, then

$$\partial_t \varphi = \tan(\vartheta_\omega(\alpha_\varphi) - \widehat{\theta})$$

exists smoothly for all time. A smooth C -subsolution gives smooth convergence [Tak20, Theorem 1.4].

- (ii) If $0 < \theta < \pi$ and $Q_\omega(\alpha_{\varphi_0}) < \pi$ on X , then

$$\partial_t \varphi = \cot Q_\omega(\alpha_\varphi) - \cot \theta$$

exists smoothly for all time. Convergence follows from a smooth C -subsolution satisfying $Q_\omega(\alpha_\varphi) < \pi$ [FYZZ25, Theorems 1.2–1.3].

- (iii) For the original flow

$$\partial_t \varphi = \vartheta_\omega(\alpha_\varphi) - \widehat{\theta},$$

if $0 < \theta < \pi/2$ and the initial potential is hypercritical and is itself a C -subsolution, the flow exists for all time and converges smoothly [Tak20, Theorem 1.3]. Without the appropriate initial restrictions, finite-time singularities occur even in smoothly solvable classes [CJ25].

The distinction matters in Problem 7.1: its almost-calibrated initial data need not be hypercritical or a C -subsolution.

1.4. Thomas–Yau: existence, flow and a toric correspondence. On a normalized Calabi–Yau manifold (Y, ω_Y, Y) , a graded Lagrangian Λ has a real phase β_Λ with

$$\omega_Y|_\Lambda = 0, \quad Y|_\Lambda = e^{\sqrt{-1}\beta_\Lambda} \text{vol}_\Lambda.$$

It is special when β_Λ is constant. Almost calibrated means $\text{osc } \beta_\Lambda < \pi$. Write $Z(\Lambda) = \int_\Lambda Y = |Z(\Lambda)|e^{\sqrt{-1}\tau(\Lambda)}$ with the phase lift in its phase interval, and normalize $\tau(\Lambda) = 0$. For

Lagrangian surfaces, flow stability requires, for every graded Hamiltonian connected-sum decomposition into almost-calibrated pieces, $\Lambda \simeq_{\text{Ham}} \Lambda_1 \# \Lambda_2$, that

$$[\tau(\Lambda_1), \tau(\Lambda_2)] \not\subset (\inf \beta_\Lambda, \sup \beta_\Lambda) \quad \text{or} \\ \text{Area}(\Lambda) < |Z(\Lambda_1)| + |Z(\Lambda_2)|.$$

The connected sum is Lagrangian neck surgery with gradings extending over the necks. The Thomas–Yau existence question seeks phase-stability conditions characterizing a special representative in a Hamiltonian isotopy class [TY02, Section 1]. It must be separated from a smooth-flow assertion:

- Neves proves that, on a compact Calabi–Yau surface, every compact embedded Lagrangian has a Hamiltonian-isotopic embedded representative whose mean curvature flow becomes singular in finite time. This applies even when the starting class contains a special Lagrangian. His relevant examples are not almost calibrated [Nev13, Theorem 6.1 and Remark 1.3].
- Lotay–Oliveira prove smooth long-time existence and convergence to a special Lagrangian for compact, embedded, circle-invariant, almost-calibrated and flow-stable Lagrangians in Gibbons–Hawking ALE/ALF hyperkähler four-manifolds [LO24, Theorem 1.2].

Section 2 distinguishes this condition from the historical Thomas–Yau quantifiers and from Joyce’s programme of categorical representatives and flow with surgery [Joy15].

A toric manifold is *weak Fano* when $-K_X$ is nef and big. For the toric correspondence, let X be a smooth projective toric weak Fano n -fold, ω a torus-invariant Kähler form, and M a line bundle with ample dual. Let $\mathcal{L}_k$ denote the SYZ transform of $M^{\otimes k}$ over the interior of the moment polytope of $(X, k\omega)$: it is the Lagrangian section determined by a Hermitian metric on that line bundle. Write $(\mathcal{Y}_{q_k}, W_k)$ for the toric Landau–Ginzburg mirror, and set

$$\mathcal{Y}_{q_k} \cong (\mathbb{C}^*)^n, \quad Y_0 = \bigwedge_{j=1}^n \frac{dz^j}{z^j}, \quad P_k(\mathcal{E}) = \int_{\gamma(\mathcal{E})} e^{-W_k} Y_0.$$

Here $\gamma(\mathcal{E})$ is the rapid-decay cycle of a Fukaya–Seidel object; its ends lie where $\text{Re } W_k \rightarrow +\infty$. A shift by r multiplies this period by $(-1)^r$.

Theorem 1.6 (Stoppa, preprint [Sto25, Theorem 1.2 and Corollary 2.7]). *Let X be a smooth projective toric weak Fano n -fold, $n \geq 2$, $\omega \in \Omega$ torus invariant, and $M^\vee$ ample. Put $A = c_1(M^\vee)$ and choose a compatible $\hat{\theta} \in ((n-2)\pi/2, n\pi/2)$, with $\theta = n\pi/2 - \hat{\theta}$. Assume*

$$\langle \mathcal{F}_p(A, \Omega; \theta), [V] \rangle \neq 0 \quad \text{for every proper toric } V, \quad 1 \leq p = \dim V < n.$$

For a sufficiently large integer k , let $\mathcal{L}_k$ be the SYZ section associated with $M^{\otimes k}$ in the mirror of $(X, k\omega)$. There are test objects $\mathcal{E}_{V,k}$, indexed by proper toric subvarieties V , and morphisms

$$\mathcal{E}_{V,k}[-r] \longrightarrow \mathcal{L}_k, \quad r = \text{codim}_X V,$$

such that $\mathcal{L}_k$ has a supercritical special SYZ representative in its Hamiltonian class if and only if

$$\text{Im} \frac{(-1)^r P_k(\mathcal{E}_{V,k})}{P_k(\mathcal{L}_k)} < 0 \quad \text{for every such } V.$$

The statement is from the cited v2 preprint. Special SYZ sections use the geometric mirror volume form, not the oscillatory form defining P_k . The theorem uses one specified family of test objects, not every subobject in an arbitrary heart. In this setting, its period test is equivalent to smooth supercritical dHYM solvability on $M^\vee$; Section 6 gives the signs and scaling [Sto25, Section 2].

1.5. Surface stability does not imply smooth solvability. On a smooth projective surface and for an ample $\Xi \in \text{NS}(X)_\mathbb{R}$, the divisorial stability condition $\sigma_{0,\Xi}$ has central charge

$$Z_{0,\Xi}(E) = -\langle e^{-\sqrt{-1}\Xi} \text{ch}(E), [X] \rangle$$

and heart obtained by tilting $\text{Coh}(X)$ at slope zero. More explicitly, let $\mathcal{T}_{0,\Xi}$ be generated by torsion sheaves and slope-semistable torsion-free sheaves of positive slope, and $\mathcal{F}_{0,\Xi}$ by slope-semistable torsion-free sheaves of nonpositive slope, where $\mu_\Xi(F) = \langle c_1(F) \smile \Xi, [X] \rangle / \text{rk}(F)$. The heart is the extension closure $\mathcal{A}_{0,\Xi} = \langle \mathcal{F}_{0,\Xi}[1], \mathcal{T}_{0,\Xi} \rangle$. Stability means

$$\text{Arg } Z_{0,\Xi}(G) < \text{Arg } Z_{0,\Xi}(E) \quad (0 \neq G \subsetneq E \text{ in } \mathcal{A}_{0,\Xi}), \quad \text{Arg} \in (0, \pi].$$

Theorem 1.7 (Mäkelä, preprint [Mak26, Corollary 1.5]). *Let X be a smooth projective surface, $\Omega \in \text{NS}(X)_\mathbb{R}$ an ample class, and L a line bundle. Put $A = c_1(L)$ and assume $s = \langle A \smile \Omega, [X] \rangle > 0$. Set*

$$A_c = A + \frac{\langle \Omega^2 - A^2, [X] \rangle}{2s} \Omega.$$

The following are equivalent:

- (i) $\langle A_c, [C] \rangle \geq 0$ for every integral curve $C \subset X$.
- (ii) $L^{\otimes k}$ is $\sigma_{0,k\Omega}$ -stable for every integer $k \geq 1$.
- (iii) $L^{\otimes k}$ is $\sigma_{0,k\Omega}$ -stable for all sufficiently large integers k .

This is the statement of the August 2026 preprint [Mak26]. The non-strict curve inequalities differ from the strict inequalities in Theorem 1.1. In particular, on

$$X = \text{Bl}_p \mathbb{P}^2, \quad \Omega = \frac{2H - E}{\sqrt{3}}, \quad L = \mathcal{O}_X(H),$$

where H is the pullback hyperplane class and E is the exceptional divisor, one has $A_c = H$, which is nef but not Kähler. Consequently,

$$\left. \begin{array}{l} L^{\otimes k} \text{ is } \sigma_{0,k\Omega}\text{-stable} \\ \text{for every integer } k \geq 1 \end{array} \right\} \text{ but no smooth dHYM solution exists.}$$

The calculation is given in Section 8 [Fan26, Example 2.11].

At a fixed scale, the same preprint [Mak26, Theorem 1.1(a)] gives the following detector. Let X be a smooth projective surface, $\Omega \in \text{NS}(X)_{\mathbb{R}}$ ample, and L a line bundle with $s = \Omega \cdot c_1(L) > 0$. Then

$$\begin{aligned} L \text{ is not } \sigma_{0,\Omega}\text{-stable} &\iff \exists D \neq 0 \text{ effective with} \\ D^2 < 0, \quad 0 < \Omega \cdot D < s, \quad \text{Arg } Z_{0,\Omega}(L(-D)) &\geq \text{Arg } Z_{0,\Omega}(L). \end{aligned} \quad (1.11)$$

Here D need not be reduced or irreducible. The full B -twisted statement and the stronger integral-curve question are separated in Theorem 8.1 and Problem 8.2.

2. THE THOMAS–YAU PROBLEM

2.1. Phase, calibration and existence. Let (Y, ω_Y, J_Y, Y) be a Calabi–Yau n -fold with Ricci-flat metric $g_Y = \omega_Y(\cdot, J_Y \cdot)$ and parallel holomorphic volume form Y , normalized so that its restriction to any oriented Lagrangian plane has unit norm. A compact oriented Lagrangian $\Lambda \subset Y$ satisfies

$$\omega_Y|_{\Lambda} = 0, \quad Y|_{\Lambda} = e^{\sqrt{-1}\beta_{\Lambda}} \text{vol}_{\Lambda}.$$

A *grading* is a continuous real lift β_{Λ} of this phase; its existence is the zero-Maslov condition. The Lagrangian is *special* when the lift is constant. It is *almost calibrated* when

$$\sup_{\Lambda} \beta_{\Lambda} - \inf_{\Lambda} \beta_{\Lambda} < \pi.$$

In that case

$$Z(\Lambda) := \int_{\Lambda} Y \neq 0, \quad Z(\Lambda) = |Z(\Lambda)|e^{\sqrt{-1}\tau(\Lambda)}, \quad \tau(\Lambda) \in [\inf_{\Lambda} \beta_{\Lambda}, \sup_{\Lambda} \beta_{\Lambda}].$$

This specifies the real lift of the cohomological phase [TY02, Sections 1–2].

Write $\Lambda' \simeq_{\text{Ham}} \Lambda$ if they are connected by an isotopy F_s for which $F_s^*(\iota_{\partial_s F_s} \omega_Y)$ is exact. The period Z is constant along such an isotopy. If $Z(\Lambda) = |Z(\Lambda)|e^{\sqrt{-1}\tau}$, then

$$\text{Vol}(\Lambda') - |Z(\Lambda)| = \int_{\Lambda'} (1 - \cos(\beta_{\Lambda'} - \tau)) \text{vol}_{\Lambda'} \geq 0. \quad (2.1)$$

Equality holds precisely when Λ' is calibrated by $\text{Re}(e^{-\sqrt{-1}\tau} Y)$. This lower bound does not supply compactness of smooth minimizers.

Problem 2.1 (Thomas–Yau [TY02, Section 1]). Let (Y, ω_Y, J_Y, Y) be a normalized Calabi–Yau n -fold and let $\Lambda \subset Y$ be a compact graded Lagrangian with $Z(\Lambda) \neq 0$. Choose $\tau \in \mathbb{R}$ with $e^{-\sqrt{-1}\tau} Z(\Lambda) > 0$. Determine phase-stability conditions on the allowed decompositions of Λ that characterize the existence of a smooth representative $\Lambda_* \simeq_{\text{Ham}} \Lambda$ satisfying

$$Y|_{\Lambda_*} = e^{\sqrt{-1}\tau} \text{vol}_{\Lambda_*}.$$

This is the existence question underlying the Thomas–Yau programme, not a definition of a universal stability condition. The choice of allowed decompositions, and whether one seeks a smooth representative or a singular categorical representative, are part of its formulation.

For the local graphical calculation, take $Y = U \times \mathbb{R}^n \subset \mathbb{C}^n$, with $z^j = x^j + \sqrt{-1}y^j$, $\omega_Y = \sum_j dx^j \wedge dy^j$ and $Y = \bigwedge_j dz^j$. For $u \in C^\infty(U, \mathbb{R})$, the graph $F_u(x) = (x, Du(x))$ satisfies

$$\begin{aligned} F_u^* \omega_Y &= \sum_{j,k} u_{jk} dx^j \wedge dx^k = 0, \\ F_u^* Y &= \det(I + \sqrt{-1}D^2u) dx^1 \wedge \cdots \wedge dx^n, \\ \beta_{F_u} &= \sum_{j=1}^n \arctan \lambda_j(D^2u). \end{aligned} \tag{2.2}$$

The scalar dHYM operator replaces this real Hessian by a Hermitian $(1, 1)$ -form. The common determinant explains the phase analogy; a global mirror correspondence requires additional geometry.

2.2. Initial conditions and mean curvature flow. For a smooth Lagrangian mean curvature flow $\partial_t F = H$, with mean curvature vector H , one has

$$H = J_Y \nabla \beta, \quad \partial_t \beta = \Delta_{\Lambda_t} \beta, \quad \frac{d}{dt} \text{Vol}(\Lambda_t) = - \int_{\Lambda_t} |H|^2 \text{vol}_{\Lambda_t}, \tag{2.3}$$

where $\Delta = \text{div} \nabla$ [TY02, Section 2]. Thus phase extrema move inward and volume decreases for as long as the flow is smooth. Neither identity controls curvature by itself.

A *graded connected sum* $\Lambda_1 \# \Lambda_2$ is an ordered Lagrangian neck surgery at intersection points, with the gradings of the two pieces extending over the necks. For the historical Thomas–Yau conditions, the allowed decompositions include the relative versions in which surgery is performed in a family [TY02, Sections 3 and 7]. With the phase lifts of that construction, write, for a decomposition $D = (\Lambda_1, \Lambda_2)$ of Λ ,

$$A_D : [\tau(\Lambda_1), \tau(\Lambda_2)] \not\subset (\inf_{\Lambda} \beta_{\Lambda}, \sup_{\Lambda} \beta_{\Lambda}), \tag{2.4}$$

$$B_D^{\leq} : \text{Vol}(\Lambda) \leq |Z(\Lambda_1)| + |Z(\Lambda_2)|. \tag{2.5}$$

After normalizing $\tau(\Lambda) = 0$, the original Conjecture 7.3 of [TY02] predicts smooth long-time existence and convergence to a special representative under

$$(\forall D, A_D) \quad \text{or} \quad (\forall D, B_D^{\leq}).$$

These are restrictions on the initial representative, not merely the assertion that its Hamiltonian class contains a special Lagrangian. The displayed conditions record the original formulation; the theorem below uses a precise almost-calibrated variant.

Theorem 2.2 (Neves [Nev13, Theorem 6.1]). *Let Y be a compact Calabi–Yau surface and let $\Sigma \subset Y$ be a compact embedded Lagrangian. There exists a compact embedded Lagrangian $\Lambda \simeq_{\text{Ham}} \Sigma$ whose mean curvature flow develops a singularity at a finite time.*

Taking Σ special Lagrangian shows that existence of a calibrated representative is insufficient for arbitrary initial data. These examples are not almost calibrated [Nev13, Remark 1.3(4)]; they do not refute the flow statement with its additional initial conditions.

Definition 2.3. Let Λ be a compact almost-calibrated Lagrangian in a Calabi–Yau surface. Rotate Y so that $\tau(\Lambda) = 0$. It is *flow stable* in the sense of [LO24, Definition 6.2] if, for every graded decomposition $\Lambda \simeq_{\text{Ham}} \Lambda_1 \# \Lambda_2$ into almost-calibrated Lagrangians, at least one of the following holds:

- $[\tau(\Lambda_1), \tau(\Lambda_2)] \not\subset (\inf_{\Lambda} \beta_{\Lambda}, \sup_{\Lambda} \beta_{\Lambda})$;
- $\text{Area}(\Lambda) < |Z(\Lambda_1)| + |Z(\Lambda_2)|$.

The disjunction is imposed separately on each decomposition, and the area inequality is strict. Neither convention should be replaced by the quantifiers or weak inequality of the original formulation.

Theorem 2.4 (Lotay–Oliveira [LO24, Theorem 1.2]). *Let Y be an ALE or ALF hyperkähler four-manifold obtained from the Gibbons–Hawking construction. Fix one of its compatible Calabi–Yau structures. Suppose $\Lambda \subset Y$ is*

- *compact, embedded and invariant under the Gibbons–Hawking circle action;*
- *almost calibrated and flow stable in the sense of Definition 2.3.*

Then its Lagrangian mean curvature flow exists smoothly for all $t \geq 0$ and converges smoothly to a circle-invariant special Lagrangian in its Hamiltonian class.

This is a result in a specified noncompact ambient geometry with symmetry, not a theorem for arbitrary Calabi–Yau surfaces.

2.3. Categorical representatives and surgery. Joyce’s formulation uses an enlarged derived Fukaya category with objects represented by branes (Λ, E, b) : graded Lagrangians, rank-one local systems E , and Floer bounding cochains b removing obstructions. Its proposed central charge is

$$Z([\Lambda, E, b]) = \int_{\Lambda} Y.$$

The expected relation is between semistability and constant-phase representatives in an appropriately enlarged class, which can include immersed or singular Lagrangians [Joy15, Conjecture 3.2].

The flow programme starts with generic Floer-unobstructed branes and allows surgeries. At regular times, it preserves the object up to isomorphism in the category; the proposed limit is assembled from special Lagrangian integral currents with ordered phases [Joy15, Conjecture 3.34]. It does not assert that topology is preserved through surgery, or that every limit is a smooth embedding. Hamiltonian isotopy, categorical isomorphism and equality of homology classes are different equivalence relations. The toric result in Section 6 concerns special SYZ sections; the scalar flow question in Section 7 concerns smooth potentials. Neither is the entire surgery programme.

3. THE SURFACE CRITERION

Let X be a compact Kähler surface. Set

$$v = \langle \Omega^2, [X] \rangle, \quad a = \langle A^2, [X] \rangle, \quad s = \langle A \smile \Omega, [X] \rangle.$$

Assume $s > 0$. Then $Z_X(A, \Omega) = v - a + 2\sqrt{-1}s$ lies in the upper half-plane. Set

$$c = \frac{v - a}{2s}, \quad \hat{\theta} = \operatorname{arccot} c \in (0, \pi), \quad A_c = A + c\Omega. \quad (3.1)$$

Theorem 1.1 (recalled). *Let X be a compact Kähler surface, $\Omega \in \mathcal{K}_X$ and $A \in H^{1,1}(X, \mathbb{R})$ with $\langle A \smile \Omega, [X] \rangle > 0$. Set*

$$\hat{\theta} = \operatorname{Arg}((\Omega + \sqrt{-1}A)^2, [X]) \in (0, \pi), \quad c = \cot \hat{\theta}.$$

The following are equivalent:

- (i) *For some, equivalently every, Kähler form $\omega \in \Omega$, there is a smooth closed real $(1, 1)$ -form $\eta \in A$ with $\vartheta_\omega(\eta) = \hat{\theta}$.*
- (ii) *$A + c\Omega \in \mathcal{K}_X$.*
- (iii) *$\langle A + c\Omega, [C] \rangle > 0$ for every irreducible curve $C \subset X$.*

Proof. Expanding the phase equation and completing the square gives

$$\begin{aligned} \alpha_\varphi^2 + 2c \alpha_\varphi \wedge \omega &= \omega^2, \\ \beta_\varphi &:= \alpha_\varphi + c\omega, \quad [\beta_\varphi] = A_c, \quad \beta_\varphi^2 = (1 + c^2)\omega^2. \end{aligned} \quad (3.2)$$

At a positive-phase solution, the eigenvalues of α_φ satisfy

$$\lambda_j + c = \frac{1 + \lambda_j^2}{\lambda_1 + \lambda_2} > 0, \quad j = 1, 2.$$

Thus $\beta_\varphi > 0$, proving (i) $\Rightarrow$ (ii).

Conversely, suppose $A_c \in \mathcal{K}_X$ and choose any Kähler form $\tilde{\omega} \in \Omega$. The identity $\langle A_c^2, [X] \rangle = (1 + c^2)v$ and Yau's theorem [Yau78] give a Kähler form $\tilde{\beta} \in A_c$ such that

$$\tilde{\beta}^2 = (1 + c^2)\tilde{\omega}^2.$$

Set $\tilde{\eta} = \tilde{\beta} - c\tilde{\omega} \in A$. Its eigenvalues λ_1, λ_2 relative to $\tilde{\omega}$ satisfy

$$(\lambda_1 + c)(\lambda_2 + c) = 1 + c^2, \quad \lambda_1 + \lambda_2 \geq 2\sqrt{1 + c^2} - 2c > 0.$$

Thus $\vartheta_{\tilde{\omega}}(\tilde{\eta}) \in (0, \pi)$ and $\cot \vartheta_{\tilde{\omega}}(\tilde{\eta}) = c$; hence $\vartheta_{\tilde{\omega}}(\tilde{\eta}) = \hat{\theta}$.

By the Hodge index theorem, $av \leq s^2$. Hence

$$\langle A_c^2, [X] \rangle = (1 + c^2)v > 0, \quad \langle A_c \smile \Omega, [X] \rangle = s + cv \geq \frac{s^2 + v^2}{2s} > 0.$$

Thus A_c lies in the positive-cone component containing Ω . The surface Kähler-cone criterion [DP04] gives (ii) $\Leftrightarrow$ (iii). $\square$

For fixed ω , the Monge–Ampère equation also gives uniqueness of the potential modulo constants.

A target is *hypercritical* when $(n-1)\pi/2 < \hat{\theta} < n\pi/2$. If the Kähler class is not prescribed, there is an elementary construction in every dimension:

$$\eta \in A, \quad \eta > 0, \quad \omega = \frac{\eta}{\tan(\hat{\theta}/n)} \implies \vartheta_\omega(\eta) = \hat{\theta}.$$

Conversely, a hypercritical solution has $\eta > 0$, since a nonpositive eigenvalue would force the phase below $(n-1)\pi/2$. This gives existence for some freely chosen background exactly when A is Kähler. It changes Ω and is not a criterion for a prescribed pair (A, Ω) .

Example 3.1. On $X = \text{Bl}_p \mathbb{P}^2$, let H be the pullback hyperplane class and E the exceptional divisor class. Take

$$\Omega = \frac{2H - E}{\sqrt{3}}, \quad A = H.$$

Then $v = a = 1$, $c = 0$ and $A_c = H$. This class is nef and big, but $\langle A_c, [E] \rangle = 0$. Thus $A_c \notin \mathcal{K}_X$, and no Kähler background $\omega \in \Omega$ admits a smooth dHYM solution.

4. SUBSOLUTIONS AND ESTIMATES

For a holomorphic line bundle L with Hermitian metric h , the Chern curvature in a local holomorphic frame e is

$$F_h = -\partial\bar{\partial} \log h(e, e), \quad c_1(L, h) = \frac{\sqrt{-1}}{2\pi} F_h.$$

If $A = c_1(L)$ and $\alpha = c_1(L, h_0)$, then

$$h = e^{-2\pi\varphi} h_0 \implies c_1(L, h) = \alpha + \sqrt{-1}\partial\bar{\partial}\varphi.$$

Thus the potential formulation includes the metric equation on a line bundle [JY17].

For a variation $v \in C^\infty(X, \mathbb{R})$, choose coordinates unitary for ω and diagonalizing α_φ . Then

$$D_\varphi(\vartheta_\omega(\alpha_\varphi))[v] = \sum_{j=1}^n \frac{v_{j\bar{j}}}{1 + \lambda_j^2}.$$

The linearization is elliptic, but its ellipticity constants degenerate when $|\lambda_j| \rightarrow \infty$.

Definition 4.1. Fix a Kähler background ω and a target phase $\hat{\theta} \in ((n-2)\pi/2, n\pi/2)$. A potential $\varphi \in C^\infty(X, \mathbb{R})$ is a C -subsolution for $\vartheta_\omega(\alpha_\varphi) = \hat{\theta}$ if the eigenvalues μ_j of $\alpha_\varphi = \alpha + \sqrt{-1}\partial\bar{\partial}\varphi$ relative to ω satisfy

$$\sum_{j \neq k} \arctan \mu_j > \hat{\theta} - \frac{\pi}{2}, \quad k = 1, \dots, n, \quad x \in X.$$

For fixed $x \in X$, the set

$$\left\{ \lambda \in \mathbb{R}^n : \lambda_j \geq \mu_j(x) \ (j = 1, \dots, n), \quad \sum_{j=1}^n \arctan \lambda_j = \hat{\theta} \right\}$$

is bounded. Indeed, if $\lambda_k \rightarrow +\infty$, then

$$\liminf_j \sum \arctan \lambda_j \geq \frac{\pi}{2} + \sum_{j \neq k} \arctan \mu_j(x) > \hat{\theta},$$

a contradiction [CJY20, Lemma 3.3]. On a surface,

$$n = 2 : \quad \mu_j + \cot \hat{\theta} > 0 \ (j = 1, 2) \iff \alpha_{\underline{\varphi}} + \cot \hat{\theta} \omega > 0.$$

Theorem 1.2 (recalled). *Let (X, ω) be a compact Kähler n -fold with $n \geq 2$, and let α be a smooth closed real $(1, 1)$ -form. Fix $\hat{\theta} \in ((n-2)\pi/2, n\pi/2)$. Assume both of the following:*

- $e^{-\sqrt{-1}\hat{\theta}} \int_X (\omega + \sqrt{-1}\alpha)^n \in \mathbb{R}_{>0}$.
- *There is a smooth C -subsolution $\underline{\varphi} \in C^\infty(X, \mathbb{R})$ for the target $\hat{\theta}$.*

Then $\vartheta_\omega(\alpha + \sqrt{-1}\partial\bar{\partial}\varphi) = \hat{\theta}$ has a smooth solution, unique modulo constants.

The published Collins–Jacob–Yau theorem assumes in addition

$$\vartheta_\omega(\alpha_{\underline{\varphi}}) > \frac{(n-2)\pi}{2} \quad [\text{CJY20, Theorem 1.2}].$$

Summing the C -subsolution inequalities gives

$$(n-1)\vartheta_\omega(\alpha_{\underline{\varphi}}) > n\left(\hat{\theta} - \frac{\pi}{2}\right).$$

Thus the extra assumption is automatic when $\hat{\theta} > (n-2+2/n)\pi/2$, but not throughout the supercritical range.

Under the Collins–Jacob–Yau hypotheses, the continuity argument uses a smoothed maximum of the initial and target phase functions, then deforms to the constant $\hat{\theta}$ [CJY20, Section 7]. For its solution potentials, normalized by $\sup_X \varphi = 0$, the estimates include

$$\begin{aligned} \|\varphi\|_{C^0(X)} &\leq C, \\ \sup_X |\sqrt{-1}\partial\bar{\partial}\varphi|_\omega &\leq C(1 + \sup_X |\nabla\varphi|_\omega^2) \quad [\text{CJY20, Proposition 3.6 and Theorem 4.1}]. \end{aligned}$$

If $M_j = \sup_X |\nabla\varphi_j|_\omega \rightarrow \infty$, rescaling at points where the maxima occur produces a limit u satisfying

$$u : \mathbb{C}^n \longrightarrow \mathbb{R}, \quad \|u\|_{L^\infty} < \infty, \quad \sqrt{-1}\partial\bar{\partial}u \geq 0, \quad |\nabla u(0)| = 1.$$

The restriction of u to each complex line is bounded and subharmonic, hence constant. This contradicts $|\nabla u(0)| = 1$ [CJY20, Proposition 5.1]. The gradient bound then gives the Hessian bound, uniform ellipticity and higher regularity.

5. NUMERICAL CRITERIA

Use $\operatorname{arccot} \lambda \in (0, \pi)$ and set

$$\theta = \frac{n\pi}{2} - \hat{\theta} \in (0, \pi), \quad Q_\omega(\eta) = \sum_{j=1}^n \operatorname{arccot} \lambda_j(\eta; \omega) = \frac{n\pi}{2} - \vartheta_\omega(\eta). \quad (5.1)$$

Then the dHYM equation is $Q_\omega(\alpha_\varphi) = \theta$. For a closed real $(1, 1)$ -form η , define

$$F_p(\eta, \omega; \theta) = \operatorname{Re}(\eta + \sqrt{-1}\omega)^p - \cot \theta \operatorname{Im}(\eta + \sqrt{-1}\omega)^p.$$

At a solution, $F_n(\alpha_\varphi, \omega; \theta) = 0$. For a complex p -plane $\Pi \subset T_x X$, $1 \leq p < n$, let $\nu_1 \leq \dots \leq \nu_p$ be the eigenvalues of $\alpha_\varphi|_\Pi$ relative to $\omega|_\Pi$. Set

$$r_\Pi = \prod_{j=1}^p \sqrt{1 + \nu_j^2}, \quad q_\Pi = \sum_{j=1}^p \operatorname{arccot} \nu_j.$$

Ordering the full eigenvalues as $\lambda_1 \leq \dots \leq \lambda_n$, eigenvalue interlacing gives

$$\nu_j \geq \lambda_j, \quad 0 < q_\Pi \leq \sum_{j=1}^p \operatorname{arccot} \lambda_j < Q_\omega(\alpha_\varphi) = \theta.$$

Hence

$$\begin{aligned} (\alpha_\varphi + \sqrt{-1}\omega)^p|_\Pi &= r_\Pi e^{\sqrt{-1}q_\Pi} \omega^p|_\Pi, \\ F_p(\alpha_\varphi, \omega; \theta)|_\Pi &= r_\Pi \frac{\sin(\theta - q_\Pi)}{\sin \theta} \omega^p|_\Pi > 0. \end{aligned}$$

Define the cohomology classes

$$\begin{aligned} \mathcal{F}_p(A, \Omega; \theta) &:= \operatorname{Re}(A + \sqrt{-1}\Omega)^p - \cot \theta \operatorname{Im}(A + \sqrt{-1}\Omega)^p \\ &= [F_p(\alpha_\varphi, \omega; \theta)] \in H^{p,p}(X, \mathbb{R}), \quad \mathcal{F}_0 = 1. \end{aligned} \tag{5.2}$$

Integration gives the necessary inequalities

$$\langle \mathcal{F}_p(A, \Omega; \theta), [V] \rangle > 0, \quad 1 \leq p = \dim V < n,$$

for every irreducible analytic subvariety $V \subset X$ [CJY20, Section 8]. On a surface,

$$\theta = \pi - \hat{\theta}, \quad \mathcal{F}_1(A, \Omega; \theta) = A + \cot(\hat{\theta})\Omega = A_c.$$

Example 5.1. Let X be a complex three-torus with no proper positive-dimensional analytic subvarieties. Take $\alpha = -\omega$, so $A = -\Omega$. The target $\hat{\theta} = 5\pi/4$, or $\theta = \pi/4$, satisfies (1.7); all proper-subvariety tests are vacuous. At a maximum of φ , however,

$$\alpha_\varphi \leq -\omega \implies \vartheta_\omega(\alpha_\varphi) \leq -\frac{3\pi}{4} \neq \frac{5\pi}{4}.$$

Thus the equation with target $\hat{\theta} = 5\pi/4$ has no solution [Zha23].

The endpoint tests do not detect this failure. Along the ray $A + t\Omega$,

$$\frac{\langle \mathcal{F}_3(A + t\Omega, \Omega; \pi/4), [X] \rangle}{\langle \Omega^3, [X] \rangle} = t^3 - 6t^2 + 6t,$$

and the value at $t = 2$ is -4 . The criterion below requires the corresponding inequalities for every $t \geq 0$.

Theorem 1.3 (recalled). *Let X be a compact Kähler n -fold with $n \geq 2$, let $A \in H^{1,1}(X, \mathbb{R})$ and let $\omega \in \Omega$ be a Kähler form. Fix $\theta \in (0, \pi)$ and assume $\langle \mathcal{F}_n(A, \Omega; \theta), [X] \rangle = 0$. Then there is a smooth closed real $(1, 1)$ -form $\eta \in A$ satisfying $Q_\omega(\eta) = \theta$ if and only if, for every $t \geq 0$ and every irreducible analytic subvariety $V \subseteq X$ of dimension $1 \leq p \leq n$,*

$$\langle \mathcal{F}_p(A + t\Omega, \Omega; \theta), [V] \rangle \begin{cases} > 0, & p < n, \\ \geq 0, & p = n. \end{cases}$$

Consequently, for compact Kähler X , solvability for one $\omega \in \Omega$ is equivalent to solvability for every such representative: all the conditions of Theorem 1.3 are cohomological.

The proof in [CLT24, Sections 3–7] constructs local potentials $u_1, \dots, u_m$ by mass concentration and regularization on resolutions, then glues them using regularized maxima. On an overlap, write $u = (u_1, \dots, u_m)$. For a smooth convex $M : \mathbb{R}^m \rightarrow \mathbb{R}$ with $\partial_i M \geq 0$ and $\sum_i \partial_i M = 1$,

$$\begin{aligned} \alpha + \sqrt{-1} \partial \bar{\partial} M(u) &= \sum_i \partial_i M(u) (\alpha + \sqrt{-1} \partial \bar{\partial} u_i) \\ &\quad + \sqrt{-1} \sum_{i,j} \partial_i \partial_j M(u) \partial u_i \wedge \bar{\partial} u_j. \end{aligned}$$

The first term is a convex combination of the local forms, and the second is semipositive. The admissible set used in [CLT24, Sections 3–7] is convex and is preserved by adding a semipositive $(1, 1)$ -form; hence the glued form remains admissible.

Theorem 1.4 (recalled). *Let X be smooth projective of dimension $n \geq 2$, let $A \in H^{1,1}(X, \mathbb{R})$ and let $\omega \in \Omega$ be a Kähler form. Fix $\theta \in (0, \pi)$. Then there is a smooth closed real $(1, 1)$ -form $\eta \in A$ satisfying $Q_\omega(\eta) = \theta$ if and only if both of the following hold:*

- $\langle \mathcal{F}_n(A, \Omega; \theta), [X] \rangle = 0$.
- $\langle \mathcal{F}_p(A, \Omega; \theta), [V] \rangle > 0$ for every irreducible proper analytic subvariety $V \subset X$ with $1 \leq p = \dim V < n$.

The additional implication in Theorem 1.4 is

$$\left. \begin{aligned} \langle \mathcal{F}_p(A, \Omega; \theta), [V] \rangle &> 0, \quad 1 \leq p = \dim V < n, \\ \langle \mathcal{F}_n(A, \Omega; \theta), [X] \rangle &= 0 \end{aligned} \right\} \implies \begin{array}{l} \text{the ray inequalities} \\ \text{of Theorem 1.3} \end{array}.$$

For a very ample integral class Γ , the relevant intersection number is the polynomial

$$\begin{aligned} &\langle \mathcal{F}_p(A + t\Gamma, \Omega; \theta), [V] \rangle \\ &= \sum_{k=0}^p \binom{p}{k} t^{p-k} \langle \mathcal{F}_k(A, \Omega; \theta) \smile \Gamma^{p-k}, [V] \rangle. \end{aligned} \tag{5.3}$$

For $0 < k < p$, a generic $(p - k)$ -fold hyperplane section represents $\Gamma^{p-k} \cap [V]$ by an effective k -cycle $\sum_\ell m_\ell [W_\ell]$. Thus

$$\langle \mathcal{F}_k(A, \Omega; \theta) \smile \Gamma^{p-k}, [V] \rangle = \sum_\ell m_\ell \langle \mathcal{F}_k(A, \Omega; \theta), [W_\ell] \rangle > 0.$$

The coefficient with $k = 0$ is $\langle \Gamma^p, [V] \rangle > 0$. The constant term is positive for $p < n$ and zero for $p = n$, by the hypotheses of Theorem 1.4. The test-family theorem of [CLT24] therefore applies to $A + t\Gamma$.

6. TORIC MIRROR CORRESPONDENCE

Let X be a smooth projective toric n -fold with $-K_X$ nef and big (a *toric weak Fano* manifold). Fix a torus-invariant Kähler form $\omega \in \Omega$ and a line bundle M with ample dual. Use

$$A = c_1(M^\vee) = -c_1(M), \quad \theta = \frac{n\pi}{2} - \hat{\theta} \in (0, \pi), \quad e^{-\sqrt{-1}\hat{\theta}} Z_X(A, \Omega) > 0.$$

The dual is essential: in the toric SYZ convention used in [Sto25, Remark 2.1], the special-section equation for M is the positive-sign dHYM equation (1.4) for $M^\vee$.

For a toric subvariety V of dimension p , put

$$T_V(M) = \int_V e^{-\sqrt{-1}\Omega} \text{ch}(M), \quad W_p(V) = \langle (A + \sqrt{-1}\Omega)^p, [V] \rangle.$$

Since $\text{ch}(M) = e^{-A}$,

$$\begin{aligned} T_V(M) &= \frac{(-1)^p}{p!} W_p(V), & W_n(X) &= R e^{\sqrt{-1}\theta}, \quad R > 0, \\ \text{Im} \frac{(-1)^{n-p} T_V(M)}{T_X(M)} &= -\frac{n! \sin \theta}{p! R} \langle \mathcal{F}_p(A, \Omega; \theta), [V] \rangle. \end{aligned} \tag{6.1}$$

Thus the numerical inequality is a strict phase inequality, with the codimension sign already fixed. For nonzero z, w , the convention

$$\text{Im}(z/w) < 0 \iff \text{Arg}_0(z/w) \in (-\pi, 0), \quad \text{Arg}_0 \in (-\pi, \pi],$$

avoids independently choosing arguments modulo 2π .

For $k > 0$, the analytic equation and its numerical tests are exactly scale invariant:

$$\begin{aligned} \mathfrak{G}_{k\omega}(k\eta) &= \mathfrak{G}_\omega(\eta), \\ \mathcal{F}_p(kA, k\Omega; \theta) &= k^p \mathcal{F}_p(A, \Omega; \theta). \end{aligned} \tag{6.2}$$

For the Landau–Ginzburg mirror $(\mathcal{Y}_{q_k}, W_k)$ of $(X, k\omega)$, let

$$\mathcal{Y}_{q_k} \cong (\mathbb{C}^*)^n, \quad Y_0 = \frac{dz^1}{z^1} \wedge \cdots \wedge \frac{dz^n}{z^n}, \quad P_k(\mathcal{E}) = \int_{\gamma(\mathcal{E})} e^{-W_k} Y_0.$$

Here $\gamma(\mathcal{E})$ is the rapid-decay integration cycle of an object in the derived Fukaya–Seidel category: its noncompact ends lie where $\text{Re } W_k \rightarrow +\infty$. A shift by r multiplies its period by $(-1)^r$. The SYZ transform of a Hermitian line bundle is a Lagrangian section over the interior of the moment polytope; changing the metric preserves its Hamiltonian class [Sto25, Section 1.1].

Theorem 1.6 (recalled). *Let X be a smooth projective toric weak Fano n -fold, $n \geq 2$, $\omega \in \Omega$ torus invariant, and $M^\vee$ ample. Put $A = c_1(M^\vee)$ and choose a compatible $\widehat{\theta} \in ((n-2)\pi/2, n\pi/2)$, with $\theta = n\pi/2 - \widehat{\theta}$. Assume*

$$\langle \mathcal{F}_p(A, \Omega; \theta), [V] \rangle \neq 0 \quad \text{for every proper toric } V, \quad 1 \leq p = \dim V < n.$$

For a sufficiently large integer k , let $\mathcal{L}_k$ be the SYZ section associated with $M^{\otimes k}$ in the mirror of $(X, k\omega)$. There are test objects $\mathcal{E}_{V,k}$, indexed by proper toric subvarieties V , and morphisms

$$\mathcal{E}_{V,k}[-r] \longrightarrow \mathcal{L}_k, \quad r = \text{codim}_X V,$$

such that $\mathcal{L}_k$ has a supercritical special SYZ representative in its Hamiltonian class if and only if

$$\text{Im} \frac{(-1)^r P_k(\mathcal{E}_{V,k})}{P_k(\mathcal{L}_k)} < 0 \quad \text{for every such } V.$$

The nonvanishing assumption is the genericity condition of [Sto25, Remark 2.6(v)]. The special-section equation uses the mirror geometric volume form; the oscillatory form $e^{-W_k} Y_0$ defines the periods in the test of Theorem 1.6. These roles should not be conflated.

The mirror comparison instead uses large-volume asymptotics of Iritani's toric $\widehat{\Gamma}$ -theorem. There are only finitely many toric tests, and their nonzero leading terms let one choose a common scale preserving every sign [Sto25, Theorem 2.5 and Section 5.2]. Together with (6.1) and Theorem 1.4, this gives the sequence of *proved equivalences in this setting*

$$\begin{array}{ccc} \text{strict numerical inequalities for } (A, \Omega) & & \\ \Downarrow & & \\ \text{smooth supercritical dHYM on } M^\vee & & \\ \Downarrow & & \\ \text{special SYZ representative of } \mathcal{L}_k & \iff & \text{the period test of Theorem 1.6.} \end{array}$$

The test objects are not asserted to exhaust all subobjects in an arbitrary categorical heart. A general Bridgeland-stability equivalence would be an additional statement.

7. PROPERNESS AND FLOWS

Fix a Kähler form $\omega \in \Omega$ and a target $\widehat{\theta}$ satisfying (1.7). Relative to this fixed target, the almost-calibrated potentials form the space

$$\mathcal{H} = \{\varphi \in C^\infty(X, \mathbb{R}) : |\vartheta_\omega(\alpha_\varphi) - \widehat{\theta}| < \pi/2 \text{ on } X\}.$$

When $\mathcal{H} \neq \emptyset$, choose α within its class so that $0 \in \mathcal{H}$. For $\varphi \in \mathcal{H}$, put

$$\begin{aligned} \rho_\varphi &= \text{Re} \left(e^{-\sqrt{-1}\widehat{\theta}} (\omega + \sqrt{-1}\alpha_\varphi)^n \right) > 0, \\ \sigma_\varphi &= \text{Im} \left(e^{-\sqrt{-1}\widehat{\theta}} (\omega + \sqrt{-1}\alpha_\varphi)^n \right). \end{aligned} \tag{7.1}$$

As in [CY18], define $\mathcal{J} : \mathcal{H} \rightarrow \mathbb{R}$ by

$$d\mathcal{J}_\varphi(v) = - \int_X v \sigma_\varphi, \quad \mathcal{J}(0) = 0.$$

For the metric

$$\langle v, w \rangle_\varphi = \int_X v w \rho_\varphi, \quad v, w \in C^\infty(X, \mathbb{R}),$$

the negative gradient flow is

$$\partial_t \varphi = -\text{grad } \mathcal{J}(\varphi) = \frac{\sigma_\varphi}{\rho_\varphi} = \tan(\vartheta_\omega(\alpha_\varphi) - \widehat{\theta}).$$

Along this flow,

$$\frac{d}{dt} \mathcal{J}(\varphi(t)) = - \int_X \tan^2(\vartheta_\omega(\alpha_\varphi) - \widehat{\theta}) \rho_\varphi. \quad (7.2)$$

Let d_1 be the path-length distance induced by $\|v\|_{1,\varphi} = \int_X |v| \rho_\varphi$. For $0 < \theta < \pi/2$, set

$$\mathcal{H}_\varepsilon = \{\varphi \in \mathcal{H} : Q_\omega(\alpha_\varphi) > \varepsilon \text{ on } X\}, \quad 0 < \varepsilon < \theta.$$

Theorem 1.5 (recalled). *Let (X, ω) be a compact Kähler n -fold with $n \geq 4$, and let α be a smooth closed real $(1, 1)$ -form. Fix $0 < \theta < \pi/2$ and $\widehat{\theta} = n\pi/2 - \theta$. Assume*

- $e^{-\sqrt{-1}\widehat{\theta}} \int_X (\omega + \sqrt{-1}\alpha)^n \in \mathbb{R}_{>0}$;
- $0 \in \mathcal{H}$.

Then $Q_\omega(\alpha_\varphi) = \theta$ is smoothly solvable if and only if, for every $\varepsilon \in (0, \theta)$, there exist $\delta_\varepsilon, C_\varepsilon > 0$ such that

$$\mathcal{J}(\varphi) \geq \delta_\varepsilon d_1(\varphi, 0) - C_\varepsilon \quad \text{for all } \varphi \in \mathcal{H}_\varepsilon \text{ with } \sup_X \varphi = 0.$$

(i) *Tangent flow.*

$$\partial_t \varphi = \tan(\vartheta_\omega(\alpha_\varphi) - \widehat{\theta}).$$

For $0 < \theta < \pi/2$, long-time existence holds from every $\varphi_0 \in \mathcal{H}$. A smooth C -subsolution gives convergence [Tak20, Theorem 1.4].

(ii) *Cotangent flow.*

$$\partial_t \varphi = \cot Q_\omega(\alpha_\varphi) - \cot \theta.$$

For $0 < \theta < \pi$, long-time existence holds when $Q_\omega(\alpha_{\varphi_0}) < \pi$ on X . Convergence follows from a smooth C -subsolution with $Q_\omega(\alpha_\varphi) < \pi$ [FYZ25, Theorems 1.2–1.3].

(iii) *The original flow.*

$$\partial_t \varphi = \vartheta_\omega(\alpha_\varphi) - \widehat{\theta}.$$

Finite-time singularities can occur even in solvable classes [CJ25]. Local convergence near a solution is known without a phase restriction [HJ20].

For the original flow with $0 < \theta < \pi/2$, a hypercritical initial potential which is itself a C -subsolution yields smooth convergence [Tak20, Theorem 1.3, citing Collins–Jacob–Yau, Remark 7.4]. There is also a symmetry-restricted global existence result: on $\text{Bl}_p \mathbb{P}^n$, $n \geq 3$, Calabi-symmetric initial and background forms with

$$\vartheta_\omega(\alpha_{\varphi_0}) > (n-2)\pi/2$$

produce a flow existing for all time [CJ25, Theorem 1.2]. This theorem does not assume smooth solvability and does not by itself assert smooth convergence.

Problem 7.1. Let (X, ω) be a compact Kähler n -fold with $n \geq 2$, let α be a smooth closed real $(1, 1)$ -form, and fix $(n-1)\pi/2 < \hat{\theta} < n\pi/2$. Assume both of the following:

- $\vartheta_\omega(\alpha + \sqrt{-1}\partial\bar{\partial}\varphi_*) = \hat{\theta}$ for some $\varphi_* \in C^\infty(X, \mathbb{R})$.
- The initial potential $\varphi_0 \in C^\infty(X, \mathbb{R})$ satisfies $|\vartheta_\omega(\alpha + \sqrt{-1}\partial\bar{\partial}\varphi_0) - \hat{\theta}| < \pi/2$ on X .

Does the evolution

$$\partial_t \varphi = \vartheta_\omega(\alpha + \sqrt{-1}\partial\bar{\partial}\varphi) - \hat{\theta}, \quad \varphi(0) = \varphi_0,$$

exist for all time and converge smoothly modulo constants?

This question isolates the initial-data issue raised after [Tak20, Theorem 1.3]; it is not the Thomas–Yau conjecture itself. The tangent-flow analogue follows from [Tak20, Theorem 1.4], since a smooth solution is a C -subsolution. Moreover,

$$\hat{\theta} > \frac{(n-1)\pi}{2}, \quad \varphi_0 \in \mathcal{H} \implies \vartheta_\omega(\alpha_{\varphi_0}) > \hat{\theta} - \frac{\pi}{2} > \frac{(n-2)\pi}{2}.$$

The hypotheses therefore lie inside the supercritical region, but do not require the initial potential to be hypercritical or a C -subsolution. The cited local, symmetric and singularity results do not prove or disprove this precise general statement.

8. SURFACE STABILITY

Let X be a smooth projective surface, let $\Omega \in \text{NS}(X)_\mathbb{R}$ be ample, and let $A = c_1(L)$ for a line bundle L . Write

$$v = \langle \Omega^2, [X] \rangle, \quad a = \langle A^2, [X] \rangle, \quad s = \langle A \smile \Omega, [X] \rangle > 0, \quad c = \frac{v-a}{2s}, \quad A_c = A + c\Omega.$$

For $B \in \text{NS}(X)_\mathbb{R}$, define the twisted slope of a torsion-free sheaf F by

$$\mu_{B,\Omega}(F) = \frac{\langle (c_1(F) - \text{rk}(F)B) \smile \Omega, [X] \rangle}{\text{rk}(F)}.$$

Let $\mathcal{T}_{B,\Omega}$ be generated under extensions by torsion sheaves and torsion-free slope-semistable sheaves of positive twisted slope. Let $\mathcal{F}_{B,\Omega}$ be generated by the torsion-free slope-semistable sheaves of nonpositive twisted slope. The tilted heart is

$$\mathcal{A}_{B,\Omega} = \left\{ E \in D^b(X) : \begin{array}{ll} H^{-1}(E) \in \mathcal{F}_{B,\Omega}, & H^0(E) \in \mathcal{T}_{B,\Omega}, \\ H^j(E) = 0 & (j \neq -1, 0) \end{array} \right\}.$$

The divisorial stability condition $\sigma_{B,\Omega}$ uses this heart and the central charge

$$Z_{B,\Omega}(E) = -\langle e^{-(B+\sqrt{-1}\Omega)} \text{ch}(E), [X] \rangle \quad [\text{Mak26, Section 2}].$$

For $B = 0$, in particular,

$$Z_{0,\Omega}(L) = \frac{v-a}{2} + \sqrt{-1}s = \frac{1}{2}Z_X(A, \Omega). \quad (8.1)$$

A nonzero $E \in \mathcal{A}_{B,\Omega}$ is $\sigma_{B,\Omega}$ -stable if

$$\text{Arg } Z_{B,\Omega}(G) < \text{Arg } Z_{B,\Omega}(E) \quad \text{for every } 0 \neq G \subsetneq E \text{ in } \mathcal{A}_{B,\Omega}, \quad \text{Arg} \in (0, \pi].$$

At $B = 0$, $s > 0$ gives $L \in \mathcal{A}_{0,\Omega}$ and $\text{Arg } Z_{0,\Omega}(L) = \hat{\theta}$.

For an integral curve $C \subset X$, set

$$N_C = \langle \Omega, [C] \rangle (v - a) + 2s \langle A, [C] \rangle = 2s \langle A_c, [C] \rangle. \quad (8.2)$$

By Theorem 1.1, dHYM solvability is equivalent to $N_C > 0$ for every integral curve $C \subset X$.

Theorem 1.7 (recalled). *Let X be a smooth projective surface, $\Omega \in \text{NS}(X)_{\mathbb{R}}$ an ample class, and L a line bundle. Put $A = c_1(L)$ and assume $s = \langle A \smile \Omega, [X] \rangle > 0$. Set*

$$A_c = A + \frac{\langle \Omega^2 - A^2, [X] \rangle}{2s} \Omega.$$

The following are equivalent:

- (i) $\langle A_c, [C] \rangle \geq 0$ for every integral curve $C \subset X$.
- (ii) $L^{\otimes k}$ is $\sigma_{0,k\Omega}$ -stable for every integer $k \geq 1$.
- (iii) $L^{\otimes k}$ is $\sigma_{0,k\Omega}$ -stable for all sufficiently large integers k .

Theorem 1.7 is from the August 2026 preprint [Mak26].

Simultaneous scaling gives

$$(A, \Omega) \longmapsto (kA, k\Omega), \quad \vartheta_{k\omega}(k\alpha_{\varphi}) = \vartheta_{\omega}(\alpha_{\varphi}).$$

For a divisor subobject $L^{\otimes k}(-C) \subset L^{\otimes k}$ in $\mathcal{A}_{0,k\Omega}$, expansion of the central charges gives

$$\begin{aligned} & \text{Im} \left(Z_{0,k\Omega}(L^{\otimes k}) \overline{Z_{0,k\Omega}(L^{\otimes k}(-C))} \right) \\ &= \frac{k^3}{2} \left(N_C - \frac{C^2 s}{k} \right). \end{aligned}$$

Consequently,

$$\text{Arg } Z_{0,k\Omega}(L^{\otimes k}(-C)) < \text{Arg } Z_{0,k\Omega}(L^{\otimes k}) \iff N_C - \frac{C^2 s}{k} > 0.$$

In particular,

$$C^2 < 0, \quad N_C = 0 \implies N_C - \frac{C^2 s}{k} > 0 \quad (k \geq 1).$$

Thus the strict subobject inequality can hold for every $k \geq 1$ while the dHYM inequality $N_C > 0$ fails.

For the blow-up in Example 3.1, take $L = \mathcal{O}_X(H)$. Then $N_C \geq 0$ for every integral curve and $N_E = 0$. Thus $L^{\otimes k}$ is $\sigma_{0,k\Omega}$ -stable for every integer $k \geq 1$, but $A_c = H \notin \mathcal{K}_X$ [Fan26, Example 2.11].

This is a boundary example: the numerical class is nef rather than Kähler. With $M = L^{\vee}$, both the ample-dual hypothesis and the strict genericity hypothesis of Theorem 1.6 fail. It disproves the implication from the stated scaled Bridgeland stability to smooth dHYM solvability, not every possible stability interpretation of the equation.

Theorem 8.1 (Mäkelä [Mak26, Theorem 1.1(a)]). *Let X be a smooth projective surface, let $B, \Omega \in \text{NS}(X)_{\mathbb{R}}$ with Ω ample, and let L be a line bundle with $s_B = \langle \Omega \smile (c_1(L) - B), [X] \rangle > 0$. Then L is not $\sigma_{B, \Omega}$ -stable if and only if there is a nonzero effective divisor D satisfying*

- $D^2 < 0$ and $\langle \Omega, [D] \rangle < s_B$;
- $\text{Arg } Z_{B, \Omega}(L(-D)) \geq \text{Arg } Z_{B, \Omega}(L)$.

The divisor need not be reduced or irreducible. The following stronger question is distinguished explicitly in [Mak26, Remark 1.2].

Problem 8.2. Let X be a smooth projective surface and fix $B, \Omega \in \text{NS}(X)_{\mathbb{R}}$ with Ω ample. Use the central charge

$$Z_{B, \Omega}(\mathcal{E}) = -\langle e^{-(B + \sqrt{-1}\Omega)} \text{ch}(\mathcal{E}), [X] \rangle.$$

Let L be a line bundle satisfying

- $s_B = \langle \Omega \smile (c_1(L) - B), [X] \rangle > 0$;
- L is not $\sigma_{B, \Omega}$ -stable.

Must there be an integral curve C such that

$$C^2 < 0, \quad \langle \Omega, [C] \rangle < s_B, \quad \text{Arg } Z_{B, \Omega}(L(-C)) \geq \text{Arg } Z_{B, \Omega}(L)?$$

Here Arg takes values in $(0, \pi]$.

For the effective divisor D supplied by Theorem 8.1, write $D = \sum_i m_i C_i$. Then

$$D^2 = \sum_i m_i^2 C_i^2 + 2 \sum_{i < j} m_i m_j (C_i \cdot C_j).$$

Since $C_i \cdot C_j \geq 0$ for $i \neq j$, the inequality $D^2 < 0$ gives a component C_i with

$$C_i^2 < 0, \quad 0 < \langle \Omega, [C_i] \rangle \leq \langle \Omega, [D] \rangle < s_B.$$

A component satisfying the phase inequality would answer Problem 8.2; that inequality does not follow from this intersection calculation. Thus the effective-divisor criterion settles the unqualified divisorial formulation, whereas the integral-curve question is an additional refinement, not a substitute for the existence problem in Section 2.

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